

# The Seismometer

Pupil sheet

Name: \_\_\_\_\_

Date: \_\_\_\_\_

**Safety:** I strike the table, never the box; nothing heavy or thrown. In the room we walk, never run.

## 1. What I see on the screen

The screen shows **Peak amplitude** (in g), **Shaking index** and a **seismogram** — the last ten seconds, on a fixed  $\pm 0.25$  g scale. Even in silence the line trembles by about 0.01 g: that is the sensor's own noise.

**The shaking index is not a magnitude.** Each unit up means shaking ten times stronger, but the index does not know how far away the strike happened — and the distance correction is exactly what makes a magnitude. **It measures how much it shook *here***, something akin to *intensity*, not to magnitude. I use it to compare strikes with each other, not to announce an earthquake.

**The silence rule:** ten seconds in which **nobody touches the table**. It is announced aloud and counted.

**The noise floor** (silence, 10 s) = \_\_\_\_\_ g    A peak smaller than that *cannot be declared detected*.

## 2. What I do on my own

**First I predict.** Will the box feel a footstep taken five metres away? ☐ yes ☐ no

From five metres the amplitude will be about \_\_\_\_\_ times smaller than from one metre.

because \_\_\_\_\_

**The sources, all from one metre.** Three strikes each, with silence between them; after each one I press **Save the peak**.

| Source        | 1 | 2 | 3 | Mean (g) |
|---------------|---|---|---|----------|
| Finger, light |   |   |   |          |
| Fist          |   |   |   |          |
| Footstep      |   |   |   |          |
| Door closed   |   |   |   |          |

**The same strike, three distances.** I use *the same* source — best of all a book dropped from the same height.

| Distance | 1 | 2 | 3 | Mean (g) |
|----------|---|---|---|----------|
| 1 m      |   |   |   |          |
| 2 m      |   |   |   |          |
| 5 m      |   |   |   |          |

**How many times it falls** from 1 m to 5 m: \_\_\_\_\_ : \_\_\_\_\_ = \_\_\_\_\_ times

**I draw the wave shape** for the strike at 1 m and for the one at 5 m. What changes — the height, or the duration too?

at 1 m

at 5 m

### 3. The conclusion I write

I predicted that, and the data show that \_\_\_\_\_  
 \_\_\_\_\_

The shaking index is not a magnitude because \_\_\_\_\_  
 \_\_\_\_\_

Real seismographs are put on bedrock, in basements, because \_\_\_\_\_  
 \_\_\_\_\_

#### Exit ticket

The same knock gave two different numbers. Why are both correct?

\_\_\_\_\_